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Journal Name

PAPER

Phosphine-catalyzed [3 + 2] annulation of 2-aminoacrylates with allenoates and mechanistic studies

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A convenient and efficient strategy for the synthesis of 3-pyrrolines containing an amino quaternary stereogenic center via phosphine-catalyzed formal [3 + 2] annulation of allenoates with 2-aminoacrylates is disclosed. A wide range of substrates underwent this reaction smoothly, affording a series of 3-pyrrolines in good to excellent yields; and it could be extended to a gram-scale reaction. The kinetic profiles and the calculated Methyl Anion Affinity (MAA) values were employed to explain the reactivites of different allenoates used in the reaction, which provides the deep understanding for the catalytic mechanism of this type of reaction.

Introduction

Pyrrolines and their derivatives represent an important class of heterocycles, which often exist in nature products and therapeutic compounds.[1] 3-Pyrroline-containing compounds possess many important biological activities including inhibitory activity against monoamine oxidase (MAO),[2] agonist activity at the NMDA receptor,[3] anticancer activity,[4] and protein GGTase-I inhibitors (shown in Fig. 1).[5] Owing to their pharmaceutical interest and synthetic utility, many methods have been developed for the synthesis of 3-pyrroline-containing compounds.[6] Nucleophilic phosphine organocatalzed [3+2] cycloaddition reaction has become one of important synthetic methods for the construction of five-membered-ring systems in recent years, since Lu's pioneer work on phosphine-catalyzed [3 + 2] cycloaddition of allenoates and activated alkenes or imines was reported.[7] Although phosphine-catalyzed [3 + 2] cycloaddition of allenoates and electron-deficient olefins or imines attract great interests for researchers in expanding its scope,[8] there are rare reports on [3 + 2] cycloaddition reactions using ketimines as substrates, which is probably due to their steric hinderance, intrinsic low reactivity.[9] As far as we know, there are only several reports[10] about phosphine-catalyzed [3 + 2] cycloadditions of allenoates and ketimines (shown in Scheme 1). For instance, Ye's group explored phosphine-catalyzed [3 + 2] cycloaddition of allenoates and cyclic ketimines.[10a] Ma and co-workers have successfully synthesized the fluorinated 3-pyrrolines by using phosphine-catalyzed cycloadditions of allenoates with electron-deficient olefins and imines.[10b] In this context, we have an interest in using a new type of ketoimine species in common phosphine-catalyzed annulation reactions to construct 3-pyrrolines and their derivatives.

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† Electronic Supplementary Information (ESI) available: Experimental procedures

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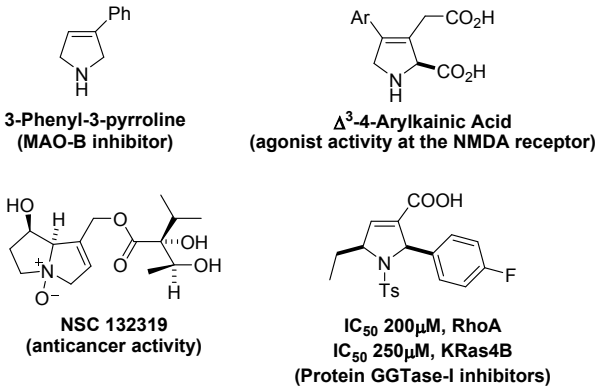
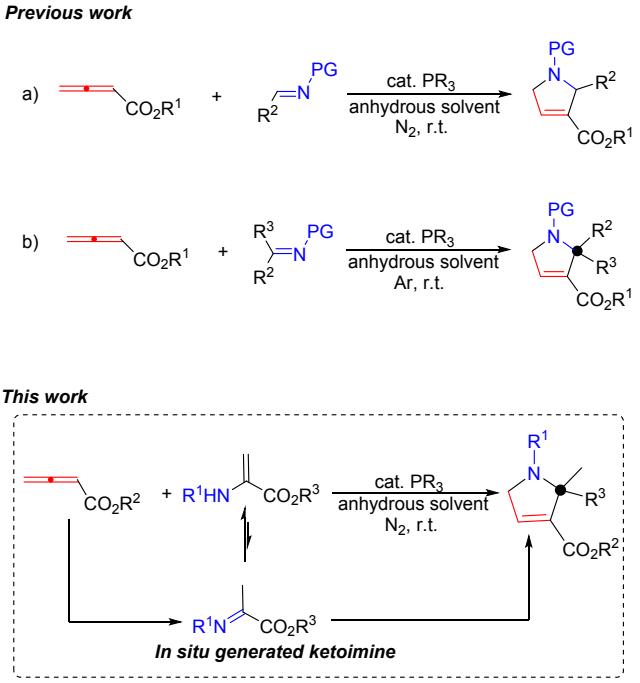


Fig. 1 Representative nature products and therapeutic molecules containing 3-pyrrolines.

In 2017, Qi's group reported an enantioselective synthesis of γ-lactam derivatives via a NHC-catalyzed [3 + 2] annulation of enals and 2-aminoacrylates.[11] Moreover, our group previously reported the synthesis of spirothioureas having an amino quaternary stereogenic center via amine-catalyzed [3 + 2] annulation of 3-isothiocyanato oxindoles and 2-aminoacrylates in 2018.[12] We assume that using such kind of in-situ generated ketoimine intermediate[13] can construct an amino quaternary stereogenic center via phosphine-catalyzed cycloaddition. Encouraged by these results and our continuous efforts on exploring the utilities of phosphine-catalyzed cycloaddition,[14] we envisaged that we could synthesize 3-pyrrolines through phosphine-catalyzed cycloaddition of 2-aminoacrylates with allenoates (Scheme 1).

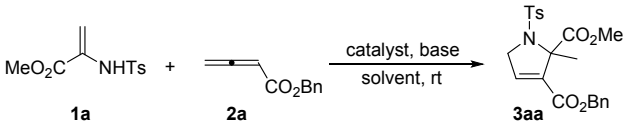


Scheme 1 Phosphine-Catalyzed [3 + 2] Cycloaddition Reactions.

Results and discussion

The phosphine-catalyzed reaction of 2-aminoacrylate **1a** with allenoate **2a** was chosen as the model system to optimize the reaction conditions. The results showed in Table 1. Initially, 2-aminoacrylate (**1a**, 1.0 equiv) was treated with allenoate (**2a**, 1.2 equiv) and PPh_3 (0.1 equiv) in 1,4-dioxane at room temperature for 4 h under an argon atmosphere. The desired product **3aa** was only obtained in trace amount (Table 1, entry 1). Subsequently, several solvents (PhMe, CH_3CN, THF, $CDCl_3$, DCM) were screened (Table 1, entries 2-6); the results showed that using THF as solvent gave the best efficiency in terms of product yield up to 92%. Furthermore, catalyst screening showed that PPh_3 was the most efficient catalyst for this reaction (Table 1, entries 7-10). Using PBu_3 as a catalyst, a cascade dimerization and cyclization of **1a** could take place.[15] Changing the ratio of **1a** and **2a** from 1:2 to 2:1 led to the production of **3aa** in a slightly decreased yield (Table 1, entries 11-12). Then, adding 2 equiv of base Cs_2CO_3 did not affect the yield significantly (Table 1, entry 13). In the absence of PPh_3, the reaction could not take place (Table 1, entry 14).

Table 1 Optimization of reaction conditions.



1a **2a** **3aa**

entry[a]	catalyst	base	solvent	yield (%)
1	PPh_3	/	1,4-dioxane	trace
2	PPh_3	/	PhMe	47
3	PPh_3	/	CH_3CN	27
4	PPh_3	/	THF	92
5	PPh_3	/	$CDCl_3$	43
6	PPh_3	/	DCM	32
7	PBu_3	/	THF	trace
8	$PPhMe_2$	/	THF	80
9	PPh_2Me	/	THF	88
10	PMe_3	/	THF	trace
11[b]	PPh_3	/	THF	82
12[c]	PPh_3	/	THF	76
13[d]	PPh_3	Cs_2CO_3	THF	56
14[e]	/	/	THF	0

[a] Unless otherwise indicated, all reactions were carried out with **1a** (0.1 mmol) **2a** (0.12 mmol) PPh_3 (10 mol%) in solvent (2.0 mL) at room temperature in a reaction tube for 4 h. [b] **1a** (0.10 mmol), **2a** (0.20 mmol). [c] **1a** (0.20 mmol), **2a** (0.10 mmol). [d] base (0.2 mmol). [e] No PPh_3 was used.

With the optimized reaction conditions in hand, we first investigated the substrate scope of this phosphine-catalyzed [3+2] annulation of allenoate **2a** with a variety of 2-aminoacrylates, and the results are summarized in Table 2. 2-aminoacrylates bearing electron-donating substituents at the phenyl ring including alkyl groups (substrates **1a**, **1b** and **1f**), alkoxy group (substrate **1c**) as well as no substituent (substrate **1e**) reacted smoothly, giving the desired products (**3aa**, **3ba**, **3fa**, **3ca**, **3ea**) in yield ranging from 70% to 92%. However, 2-aminoacrylates (substrates **1h**, **1i**, **1j**, **1k**, **1l**, **1m**, **1n**, except for **1g**) bearing electron-withdrawing substituents on the benzene ring significantly affected the reaction, leading to the yields dropping; the corresponding products **3ha**, **3ia**, **3ja**, **3ka**, **3la**, **3ma**, **3na** were accessed in moderate yields ranging from 42%

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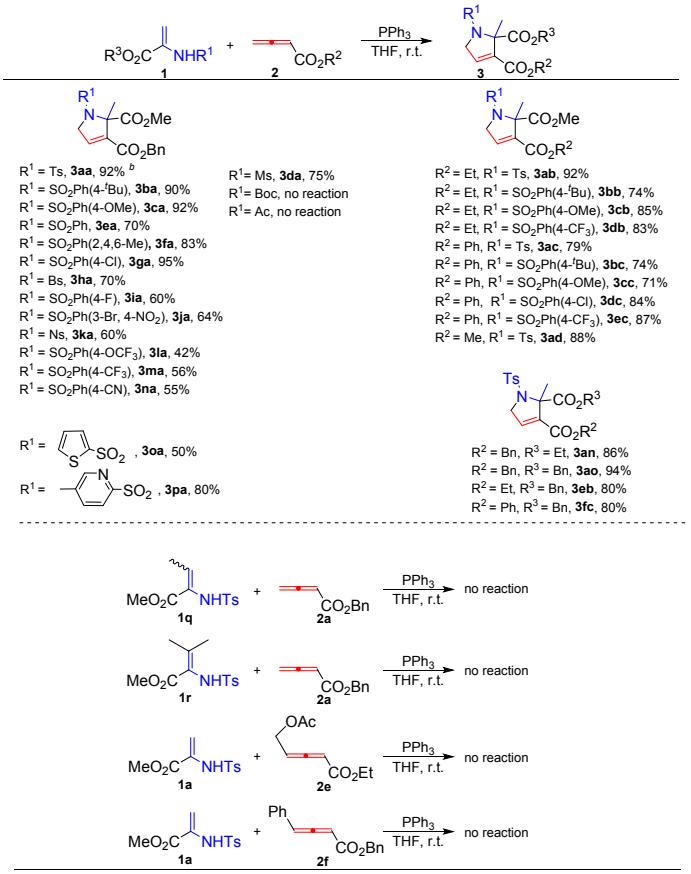
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to 70%. The substrate **1d** having methylsulfonyl group as protecting group underwent this reaction smoothly, affording the product **3da** in 75% yield. However, no reaction took place when the substrate's protecting group was replaced by acyl protecting groups such as Ac and Boc. Besides the arene sulfonyl groups, the sulfonyl groups containing heterocycle such as thiophene (substrate **1o**) and 5-methylpyridine (substrate **1p**) have been proved to be compatible, and generated the desired products (**3oa**, **3pa**) in yields of 50% and 80%, respectively. Moreover, no reactions took place when 2-aminoacrylates[16] with β-substitutions (**1q**, **1r**) were used. Subsequently, the reactions of several representative 2-aminoacrylates having sulfonyl protecting group containing methyl, butyl, alkoxy and trifluoromethyl of with other allenoates (**2b**, **2c**, **2d**) were investigated. The reactions of 2-aminoacrylates containing electron-donating groups with alkyl allenoates (substrates **2b**, **2d**) and phenyl allenoate (substrate **2c**) proceeded smoothly, affording the corresponding products **3ab**, **3bb**, **3cb**, **3ac**, **3bc**, **3cc**, **3ad**; their yields were similar to those products obtained from allenoate **2a**. The reactions of 2-aminoacrylates containing electron-withdrawing groups with allenoates (**2b**, **2c**) gave the desired products **3db**, **3ec**, whose yields increased compared with the product **3ma**. Next, in order to evaluate the effect of ester groups in substrates **1** on the reaction, the methyl ester was displaced by ethyl ester (substrate **1n**), and benzyl ester (substrate **1o**). The corresponding reaction of **1n** with **2a** gave an excellent yield of 86% for product **3an**. The substrate **1o** reacted with benzyl (substrate **2a**), alkyl (substrate **2b**) and phenyl allenoate (substrate **2c**), providing the expected products **3ao**, **3eb**, **3fc** in 80-94% yields. In addition, it is regrettable that γ-substituted allenoates (**2e, 2f**) did not react with **1a** under the standard conditions.

To further demonstrate the benefits of employing phosphine catalysis in [3 + 2] annulation, the enantioselective variant was explored (for details, see Supporting Information). A series of bifunctional chiral phosphine organocatalysts were tested for this reaction. Thiourea-derived chiral catalysis **L** was found to provide the product **3aa'** in excellent yield with moderate enantioselectivity (92% yield and 52% ee., Scheme 2). We also attempted the asymmetric reactions of **1a** with **2c**; unfortunately, an excellent enantioselectivity was not achieved (for details, see Supporting Information).

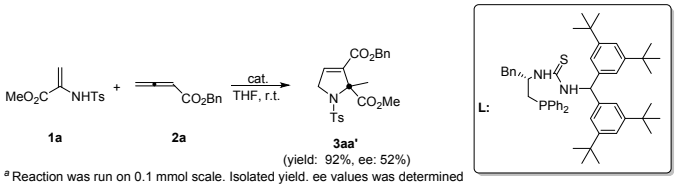
Table 2 Reaction of 2-aminoacrylate **1** with allenoates **2**.[a]

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[a] Unless otherwise indicated, all reactions were carried out with **1** (0.1 mmol), **2** (0.12 mmol) PPh_3 (10 mol%) in THF (2.0 mL) at room temperature in a reaction tube for 4 h. [b] Isolated yield.

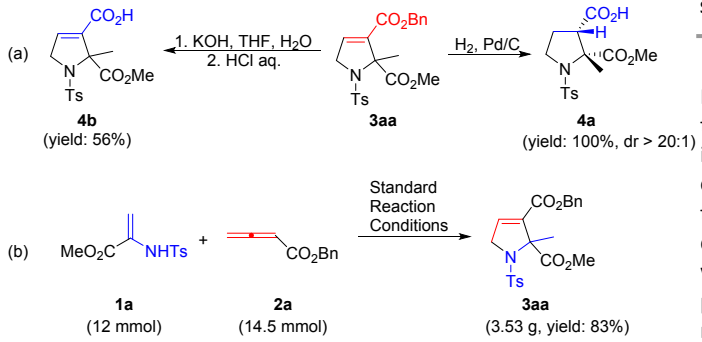
To demonstrate the synthetic utility of the 3-pyrroline products **3**, some transformations and a large-scale production were performed (Scheme 3). Hydrogenation of 3-pyrroline **3aa** could take place catalyzed by Pd/C, affording carboxylic acid **4a** in excellent yield and diastereoselectivity **(**more than 20:1 dr value). The structure of **4a** was assigned by X-ray diffraction (Fig. 2). On the other hand, in the presence of strong base, the debenzylation of **3aa** was easy to occur, producing pyrrole-fused-carboxylic acid **4b** in yield of 56%. In addition, to demonstrate the scalability of this phosphine-catalyzed [3 + 2] cycloaddition reaction, we carried out the model reaction on a 12 mmol scale to provide the expected product **3aa** in 83% yield (Scheme 3b). This demonstrated that our methodology could be potentially scaled up to gram-scale synthesis.



[a] Reaction was run on 0.1 mmol scale. Isolated yield. ee values was determined by HPLC analysis on a chiral stationary phase

Scheme 2 Asymmetric reaction of 2-aminoacrylate **1a** with allenoate **2a**.[a]

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Scheme 3 The synthetic utility of the final products and the gram-scale synthesis.

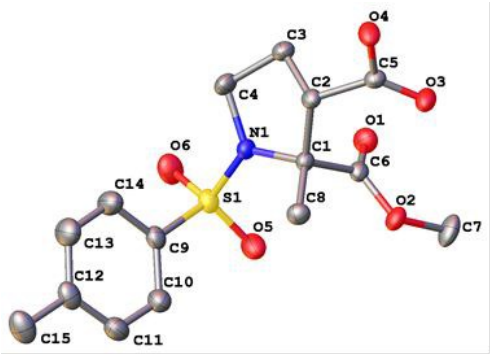
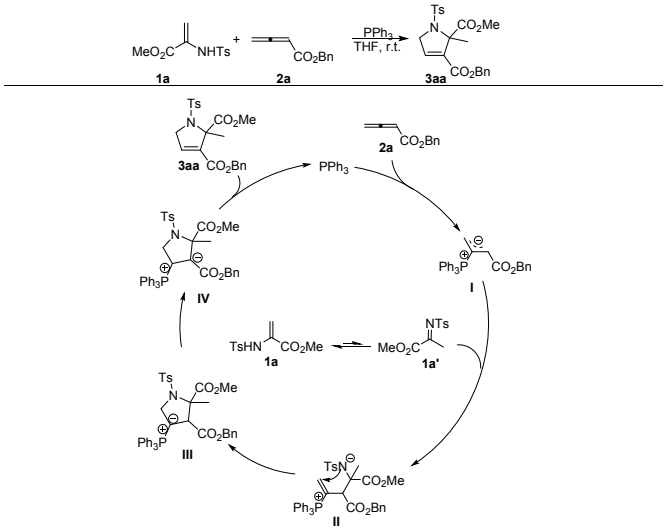


Fig. 2 Single-crystal X-ray structure of **4a**.

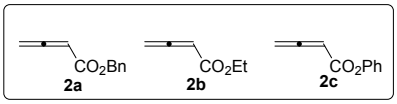
Based on the commonly accepted mechanism for phosphine-catalyzed cycloaddition reaction, the proposed reaction mechanism is shown in Scheme 4. The cycloaddition reaction is initiated by the *in situ* generated zwitterionic intermediate **I** through the addition of a tertiary phosphine to the allenoate **2a**. The intermediate **I** acts as a 1,3-dipole and undergoes a nucleophilic addition with *in situ* generated ketoimine intermediate **1a'** via tautomerization of 2-aminoacrylate **1a** to give a phosphorus ylide **II**. Next, the ylide **II** completes a cycloaddition to form an intermediate **III**, which subsequently undergoes a formal [1,2]-proton shift to yield intermediate **IV** and followed by an elimination process to afford cycloaddition product **3aa**.

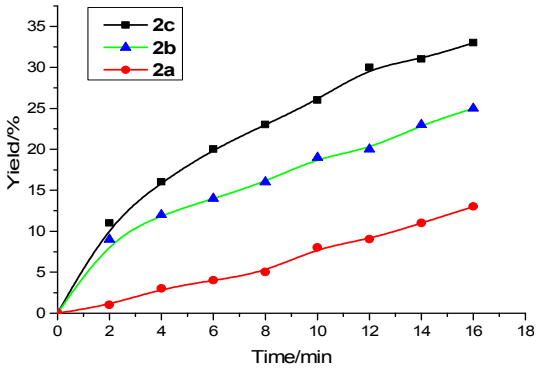


Scheme 4. Proposed Reaction Mechanism.

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In a recent report,[17] it was shown that the calculated Methyl Anion Affinity (MAA) values can serve as indicators of the electrophilicities of Michael acceptors. We would like to investigate whether there are some correlations between the calculated MAA values and the reactivities of allenoates in this type of cycloaddition reaction; thus, a series of time-dependent NMR studies were carried out. The reactions of **1a** with several allenoates **2a**, **2b** and **2c** were tested in the presence of the 10 mol% or 30 mol% of PPh_3 in $CDCl_3$, respectively. The kinetic profiles were shown in Fig. 3 and Fig. 4 (for details, see Supporting Information). The experimental results were consistent with the calculation results (Fig. 5). The benzyl 2,3-butadienoate (substrate **2a**), having the smallest MAA value, demonstrated the lowest reaction rate. The kinetic curves of ethyl 2,3-butadienoate (substrate **2b**) also fit the calculation results as well. Moreover, phenyl 2,3-butadienoate (substrate **2c**) having the largest MAA value demonstrated the fastest reaction rate among the tested allenoates. These findings indicate that the calculated MAA values can serve as indicators for the reactivities of allenoates in this cycloaddition reaction, which may help the further development for this type of reaction. It should be mentioned that the turnover curves in Figs. 3 and 4 are in agreement with the situation of "pre-steady state kinetics" often seen in multistep catalytic sequences.[18] These schemes are characterized by very fast initial product formation (boost phase) for the very first turnover cycle of the catalyst, and subsequent slowdown due to a decrease in the concentration of free catalyst. It is very nicely seen for the turnover curves for **2c**, whose first point (at 2 min) on the turnover curves in Fig. 3 and 4 correspond effectively to the catalyst concentrations. The situation is really different for allenoate **2a**. The reaction of **2a** appears to get faster at lower catalyst concentration. This interesting result may indicate that for **2a** the last step in the catalytic cycle shown in Scheme 4 is slower as compared to the other allenoates. An active intermediate could be detected by ^{31}P NMR during the reaction (for details, see Fig. S7 in SI). In summary, the MAA values can reveal that the electrophilicities of Michael acceptors effectively influence the overall reaction rate; however, the influences of other steps in the the multisetp catalytic cycles could not be ignored in some cases.





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Fig. 3 Time-dependent ^{1}H-NMR studies of the reaction in the presence of the 10 mol% of PPh_3.

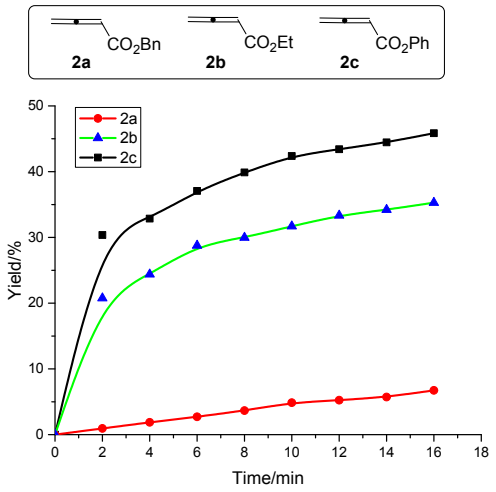


Fig. 4 Time-dependent ^{1}H-NMR studies of the reaction in the presence of the 30 mol% of PPh3.

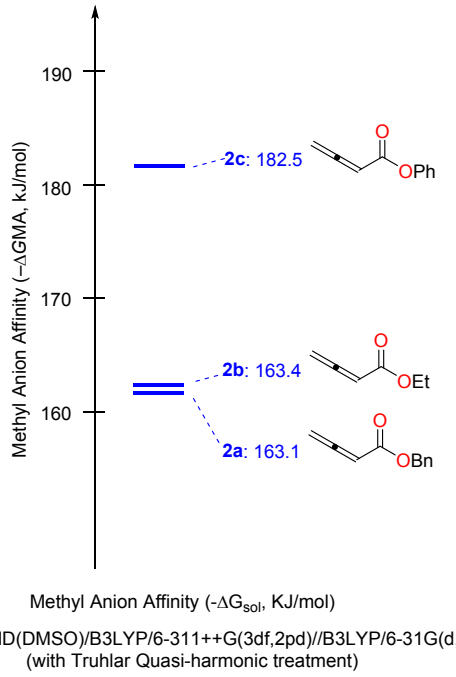


Fig. 5 Methyl Anion Affinity (MAA) values of the tested allenoates.

Conclusions

In conclusion, we have developed a novel tertiary phosphine catalyzed [3 + 2] annulation of 2-aminoacrylates with allenoates, affording the respective 3-pyrroline derivatives under mild conditions. A new type of *in situ* generated ketoimine is applied to construct an amino quaternary stereogenic center, which provides a new way of thinking about constructing the biologically and pharmacologically important 3-pyrroline derivatives. The kinetic profiles and the calculation of Methyl Anion Affinity (MAA) values reveal that different kinds of allenoates can effectively influence the reaction rate. The further studies on the asymmetric catalytic version of this reaction are currently underway in our laboratory.

Experimental

General experimental method

NMR spectra were recorded with a Bruker spectrometer at 400 MHz (^{1}H NMR), 100 MHz (^{13}C NMR) and 376 MHz (^{19}F NMR) in $CDCl_3$, respectively. Chemical shift were reported in ppm down field from internal TMS. Organic solvents used were dried by standard methods when necessary. Commercially available reagents were used without further purification. All reactions were monitored by TLC with Huanghai GF254 silica gel coated plates. Flash column chromatography was carried out using 300-400 mesh silica gel at increased pressure. Infrared spectra were recorded on a Perkin-Elmer PE-983 spectrometer with absorption in cm^{-1}. Mass spectra were recorded by ESI and HRMS was measured on a HP-5989 instrument.

Conflicts of interest

There are no conflicts of interest to declare.

Acknowledgment

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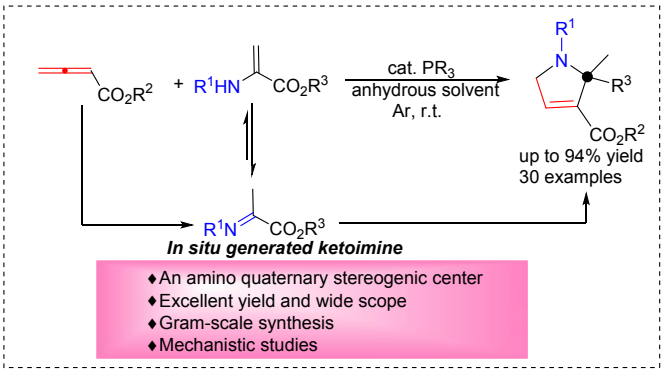
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Phosphine-catalyzed [3 + 2] annulation of 2-aminoacrylates with allenoates and mechanistic studies



A convenient and efficient strategy for the synthesis of 3-pyrrolines containing an amino quaternary stereogenic center via phosphine catalyzed formal [3 + 2] annulation of allenoates with 2-aminoacrylates is disclosed. The reaction is easy to perform in good to excellent yields and extend to a gram-scale reaction. The catalytic mechanism was investigated by DFT and kinetic studies.

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